Target-Population Fragility in Judge IV under Average Monotonicity

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Abstract

Judge-IV designs increasingly invoke average monotonicity when strong monotonicity is hard to defend. We show that average monotonicity does not preserve a stable target population: the response types receiving positive weight switch discretely when a judge's treatment propensity crosses the assignment-weighted average, so when the switching types are present in the population and have distinct average treatment effects, the estimand need not be locally invariant. We propose a diagnostic that measures each judge's standardized distance to this boundary. In the Philadelphia bail data, three of eight judges, handling about 29 percent of cases, lie near the boundary. Target-population stability is therefore an empirical object once average monotonicity is invoked.

Keywords: Judge-IV designs; Average monotonicity; Target population; Complier weights; Pretrial detention

JEL Classification: C21, C26, C18

1 Introduction

Instrumental-variable designs based on the random assignment of judges have become a standard source of quasi-experimental variation, used to estimate the effects of incarceration, pretrial detention, disability insurance, and other consequential decisions.[1] Their

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[1] Prominent applications include Kling (2006); Maestas et al. (2013); Doyle Jr et al. (2015); Dobbie et al. (2017, 2018), and more recent work such as Humphries et al. (2025); Garin et al. (2025).

interpretation rests on monotonicity. Strong monotonicity, the requirement that a judge who is stricter on average be weakly stricter in every case, is often hard to defend. Thus, a growing literature has turned to average monotonicity, a weaker condition that preserves the IV estimand as a proper weighted average of treatment effects even when judges rank cases differently (Frandsen et al., 2023; Chyn et al., 2025). That shift fixes the sign of the weights, but it leaves a separate question open. This paper asks whether the IV estimand under average monotonicity identifies the effect for a stable target population.

We answer this question in the smallest setting that displays the mechanism, a three-judge design with eight response types. Under strong monotonicity, the target population is defined by behavior, namely the individuals whose treatment changes when they face a stricter judge. Under average monotonicity, the set of positively weighted types depends instead on where each judge's propensity lies relative to the assignment-weighted average $\bar{p}$. Two non-monotone types sit at that boundary with weights of opposite sign, so when a judge's propensity crosses $\bar{p}$, one type leaves the positively weighted population and its mirror enters. If the two types are both present in the population and have distinct average treatment effects, the identified parameter need not be locally invariant, even though the propensity profile changed by an arbitrarily small amount. Average monotonicity continues to rule out negative weights, but it does not pin the estimand to a population fixed by behavior alone. Where Mogstad and Torgovitsky (2024) note that the target population identified under average monotonicity depends on the instrument distribution, we show that this dependence is discontinuous: an arbitrarily small change in the propensity profile can change which response types the estimand averages over, not only their weights. The same logic extends to any finite number of judges.

We turn this result into a boundary diagnostic for applied work and apply it to the Philadelphia bail data of Stevenson (2018). Because researchers estimate judge propensities rather than observe them, the relevant question is whether the data place each judge securely on one side of $\bar{p}$. We report each judge's estimated distance to the boundary, that distance in standard-error units, and the assignment share handled by judges close to the boundary. In Philadelphia, three of eight judges are boundary-fragile and together handle about 29 percent of cases, even though earlier work finds no decisive evidence against the design in the aggregate sample (Coulibaly et al., 2024). This does not reject the design or test average monotonicity; it shows that target-population stability should be reported alongside balance, first-stage, and exclusion checks, not as a specification test but as a measure of how sensitive the estimand's interpretation is to boundary classifications (Chyn et al., 2025; Goldsmith-Pinkham et al., 2025).

2 IV Weights under Average Monotonicity

We work in the standard judge-IV design with random assignment and exclusion, where the IV estimand is a weighted average of individual treatment effects. Let $Z_i \in \{1, \ldots, J\}$ denote the judge assigned to individual i, assigned with probability $\lambda_z > 0$, and let $D_{iz} \in \{0, 1\}$ be the treatment individual i would receive under judge z. Potential outcomes are Y_{1i} and Y_{0i}, with treatment effect $\delta_i = Y_{1i} - Y_{0i}$. We maintain that judge assignment is independent of $(Y_{1i}, Y_{0i}, \{D_{iz}\}_z)$ and affects outcomes only through treatment. Writing $p_z = E[D_i \mid Z_i = z]$ for judge z's propensity and $\bar{p} = \sum_z \lambda_z p_z$ for the assignment-weighted average propensity, the estimand is $\beta^{IV} = E[\omega_i \delta_i]/E[\omega_i]$.

The weight on individual i is the covariance, across judge assignments, between judge propensities and i's potential treatment profile, and its sign governs whether IV retains a weighted-average interpretation. This weight is

$$\omega_i = \sum_{z=1}^{J} \lambda_z (p_z - \bar{p})(D_{iz} - \bar{D}_i), \qquad \bar{D}_i = \sum_{z=1}^{J} \lambda_z D_{iz}. \tag{2.1}$$

It is positive when judges who treat more often in the population are also more likely to treat individual i. When treatment effects are heterogeneous, an individual with $\omega_i < 0$ enters β^{IV} with negative weight, and the estimand can no longer be read as an average effect for a positively weighted subpopulation.

Strong monotonicity fixes the target population through behavior, whereas average monotonicity asks only that each individual's weight be nonnegative. Strong monotonicity requires $D_{iz} \geq D_{iz'}$ whenever $p_z \geq p_{z'}$, so that a more treatment-prone judge treats everyone a less treatment-prone judge would treat. This pins the estimand to a fixed complier population but forces judges to differ along a single leniency dimension, a restriction that fails when judges rank cases differently across margins. Average monotonicity weakens this requirement to a sign condition on the individual weight.

Assumption 1 (Average Monotonicity) *For each individual i, $\omega_i \geq 0$ almost surely.*

Average monotonicity allows individual profiles that violate the common ranking of strong monotonicity while still preserving nonnegative weights, so β^{IV} remains a proper weighted average of treatment effects, as in Frandsen et al. (2023).

Average monotonicity removes negative weights but ties the admissible response types to where each judge sits relative to the average propensity, not to behavior alone (Mogstad and Torgovitsky, 2024). Describe individual i by a response type $\tau = (\tau_1, \ldots, \tau_J) \in \{0, 1\}^J$, where τ_z is the treatment status under judge z. Because $\sum_z \lambda_z(p_z - \bar{p}) = 0$, the type-specific weight is $\omega(\tau) = \sum_z \lambda_z(p_z - \bar{p})\tau_z$, and average monotonicity requires $\omega(\tau) \geq 0$ for every type that appears in the population. Whether a type clears this bar depends on the propensity profile $(p_1, \dots, p_J)$ and the assignment probabilities $(\lambda_1, \dots, \lambda_J)$, not on the response type alone. Average monotonicity therefore solves the negative-weight problem without guaranteeing that β^{IV} refers to a target population fixed by individual behavior, an observation we sharpen in the next section.

3 Boundary Discontinuity

The three-judge design is the smallest setting in which average monotonicity changes the target population without changing the sign of the IV weights. Order the judges so that $p_1 \leq p_2 \leq p_3$. The two extreme judges generate ordinary compliers; three judges is the smallest design that also admits individuals whose treatment is non-monotone in judge propensity while leaving $\bar{p}$ free to move between them.

Written relative to the intermediate judge, the weight makes the sign of ω_i depend on treatment status under the extreme judges compared with the middle judge. Since $\sum_{z=1}^{3} \lambda_z(p_z - \bar{p}) = 0$,

$$\omega_i = \lambda_1(p_1 - \bar{p})(D_{i1} - D_{i2}) + \lambda_3(p_3 - \bar{p})(D_{i3} - D_{i2}). \tag{3.1}$$

Because $p_1 \leq \bar{p} \leq p_3$, the first coefficient is nonpositive and the second nonnegative, so the sign of ω_i is set by how individual i's treatment under the lenient and strict judges compares with treatment under the intermediate judge. Each individual belongs to one of eight response types $g = (D_{i1}, D_{i2}, D_{i3})$, enumerated with their weights and admissibility under strong and average monotonicity in Appendix A.

Six of the eight response types have an admissibility that does not depend on the propensity profile. Always-takers and never-takers receive zero weight; the monotone compliers $(0, 0, 1)$ and $(0, 1, 1)$ are admissible under both conditions; the reverse types $(1, 1, 0)$ and $(1, 0, 0)$ carry negative weight and are ruled out under average monotonicity. None of these classifications turns on where p_2 lies relative to $\bar{p}$.

The two non-monotone mirror types, $g_4 = (1, 0, 1)$ and $g_8 = (0, 1, 0)$, carry weights of opposite sign, fixed entirely by the sign of $p_2 - \bar{p}$. Type g_4 is treated by the lenient and strict judges but not the intermediate one; type g_8 is treated only by the intermediate judge. Strong monotonicity rules out both, since neither profile is weakly increasing in judge propensity.

Their weights are

$$\omega(g_4) = -\lambda_2(p_2 - \bar{p}), \qquad \omega(g_8) = \lambda_2(p_2 - \bar{p}). \tag{3.2}$$

When $p_2 < \bar{p}$, average monotonicity admits g_4 and excludes g_8; when $p_2 > \bar{p}$, the roles reverse; when $p_2 = \bar{p}$, both receive zero weight.

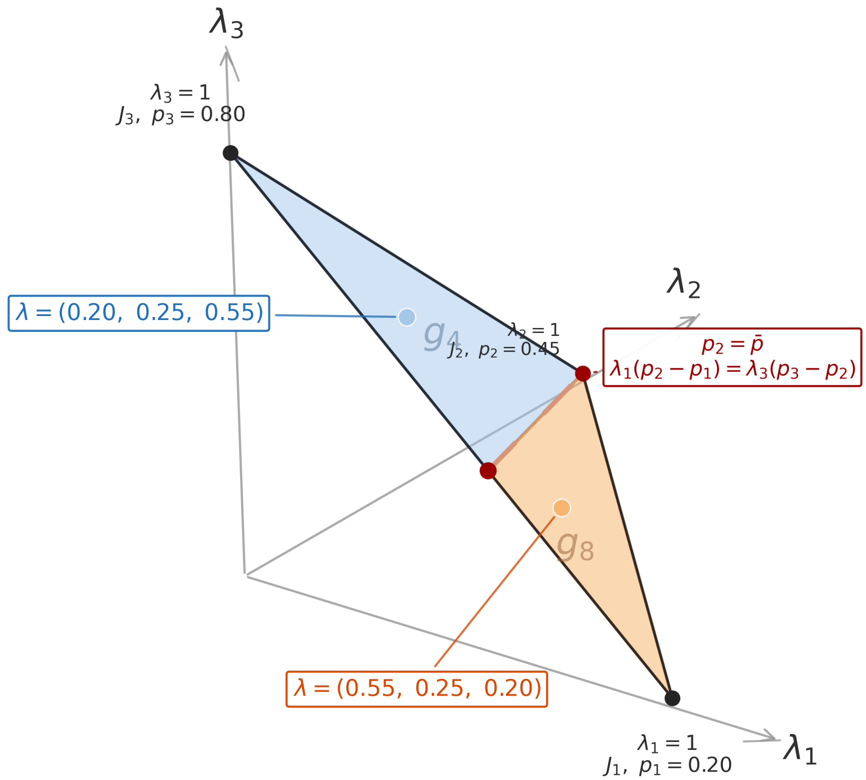

Figure 1: The Average-Monotonicity Boundary in the Assignment-Share Simplex (Three Judges)

Notes: The figure fixes judge propensities at $p_1 = 0.20$, $p_2 = 0.45$, $p_3 = 0.80$ and varies the assignment shares $(\lambda_1, \lambda_2, \lambda_3)$ over the simplex. The red locus is the average-monotonicity boundary $p_2 = \bar{p}$, equivalently $\lambda_1(p_2 - p_1) = \lambda_3(p_3 - p_2)$. In the blue region $p_2 < \bar{p}$ and the non-monotone type $g_4 = (1, 0, 1)$ receives positive weight; in the orange region $p_2 > \bar{p}$ its mirror $g_8 = (0, 1, 0)$ receives positive weight. Crossing the boundary switches the positively weighted type discretely rather than reweighting smoothly.

The admissible set therefore changes discontinuously as p_2 crosses $\bar{p}$, and when the switching types are present and have distinct average treatment effects, the estimand need not be locally invariant. All six profile-independent types keep their sign through the crossing, so the only change in the positive-weight set is the switch between g_4 and g_8: one type exits and its mirror enters, as Figure 1 shows in the assignment-share simplex.

Proposition 1 (Boundary discontinuity) *In the three-judge design with $p_1 \leq p_2 \leq p_3$, the set of response types consistent with average monotonicity changes discontinuously at*

$p_2 = \bar{p}$.

Let $\tau_g = E[\delta_i \mid \text{type } g]$. Because β^{IV} is a weighted average over the admissible types, the crossing replaces a weighted average that loads on g_4 with one that loads on g_8. If the switching types carry positive mass and $\tau_{g_4} \neq \tau_{g_8}$, the identified parameter need not be locally invariant, even though the propensity profile changed by an arbitrarily small amount.

Corollary 1 (Target-population instability) *If the switching types have positive population shares and $\tau_{g_4} \neq \tau_{g_8}$, the IV estimand under average monotonicity need not be locally invariant to perturbations of the propensity profile near $p_2 = \bar{p}$.*

The same mechanism extends to any finite number of judges. For any judge z, the type treated only by z carries weight $\lambda_z(p_z - \bar{p})$ and its componentwise complement carries $-\lambda_z(p_z - \bar{p})$, so the two switch sign exactly when p_z crosses $\bar{p}$. Appendix B states and proves the finite-J result; we use the three-judge case in the text because it displays the mechanism without the additional notation. Proofs of Proposition 1 and Corollary 1 are in Appendix A.

4 A Boundary Diagnostic in Philadelphia Bail

The boundary result becomes empirical once judge propensities are estimated, so the relevant question is whether the data place each judge securely on one side of $\bar{p}$. If a judge's estimated propensity lies close to the assignment-weighted average, the sample classifies that judge only weakly as above or below the boundary, and under average monotonicity that classification determines which mirror response types can receive positive weight. The exact crossing $p_z = \bar{p}$ is non-generic; what matters in practice is whether the sample separates each judge from the boundary.

We propose the standardized boundary distance $\hat{r}_z$ and the fragile set $\widehat{\mathcal{F}}(c)$, a sign-classification statistic rather than a test of average monotonicity. Let $\hat{p}_z$ be the estimated propensity of judge z and $\hat{\lambda}_z$ the empirical assignment share, with $\hat{\bar{p}} = \sum_j \hat{\lambda}_j \hat{p}_j$ and boundary distance $\hat{d}_z = \hat{p}_z - \hat{\bar{p}}$. The standardized distance and fragile set are

$$\hat{r}_z = \frac{|\hat{d}_z|}{\widehat{\text{se}}(\hat{d}_z)}, \qquad \widehat{\mathcal{F}}(c) = \{z : \hat{r}_z \leq c\}. \tag{4.1}$$

A judge in $\widehat{\mathcal{F}}(c)$ is one the sample does not separate from the boundary at cutoff c. The statistic does not test average monotonicity; it measures whether the sample places each judge securely on one side of the boundary that determines average-monotonicity weights.

Appendix C gives the propensity and standard-error construction, and Appendix D shows in simulations that low $\hat{r}_z$ maps into frequent sign misclassification.

We apply the diagnostic to the Philadelphia bail data of Stevenson (2018), where bail magistrates rotate across shifts and the scheduled magistrate provides quasi-random variation in pretrial detention. The outcome is conviction, the treatment is pretrial detention, and the excluded instruments are eight judge indicators, in a maximum-control specification with defendant characteristics, prior-record variables, offense indicators, and court-calendar controls over 331,971 cases. This setting is useful precisely because earlier work finds no decisive evidence against the design in the aggregate sample (Coulibaly et al., 2024), so boundary fragility can be assessed on its own terms rather than as a symptom of a failed design.

Three judges are boundary-fragile at $c = 1.96$ and together handle about 29 percent of cases. Figure 2 plots $\hat{d}_z$ with 95 percent confidence intervals against the boundary at zero. Judges 1, 5, and 7 fall inside the fragile set: Judge 7 is closest, with $\hat{d}_7 = 0.0009$, $\hat{r}_7 = 0.44$, and $\hat{\lambda}_7 = 0.125$, so the minimum standardized distance is not driven by a judge with negligible case load; Judges 1 and 5 lie just below the boundary, with $\hat{r}_1 = 1.45$ and $\hat{r}_5 = 1.69$. Such fragility is not peculiar to Philadelphia: Appendix D shows that in many-judge designs some judge typically lies near the boundary, so these distances are worth reporting routinely.

Boundary fragility is a target-population diagnostic, not a rejection of the design. A judge near the boundary does not imply that assignment is nonrandom, that exclusion fails, or that strong monotonicity is violated. It instead says that once average monotonicity is invoked, the positive-weight population is not sharply classified for judges near the boundary. When those judges carry nontrivial case mass, the sign of their contribution to β^{IV} is not pinned down by the sample, so a cautious reading attaches the estimate to a design-dependent population rather than a fixed complier group. Reporting each judge's distance $\hat{d}_z$, its standardized value $\hat{r}_z$, and the assignment share of the fragile set answers a question that balance tests, first-stage statistics, and exclusion arguments do not.

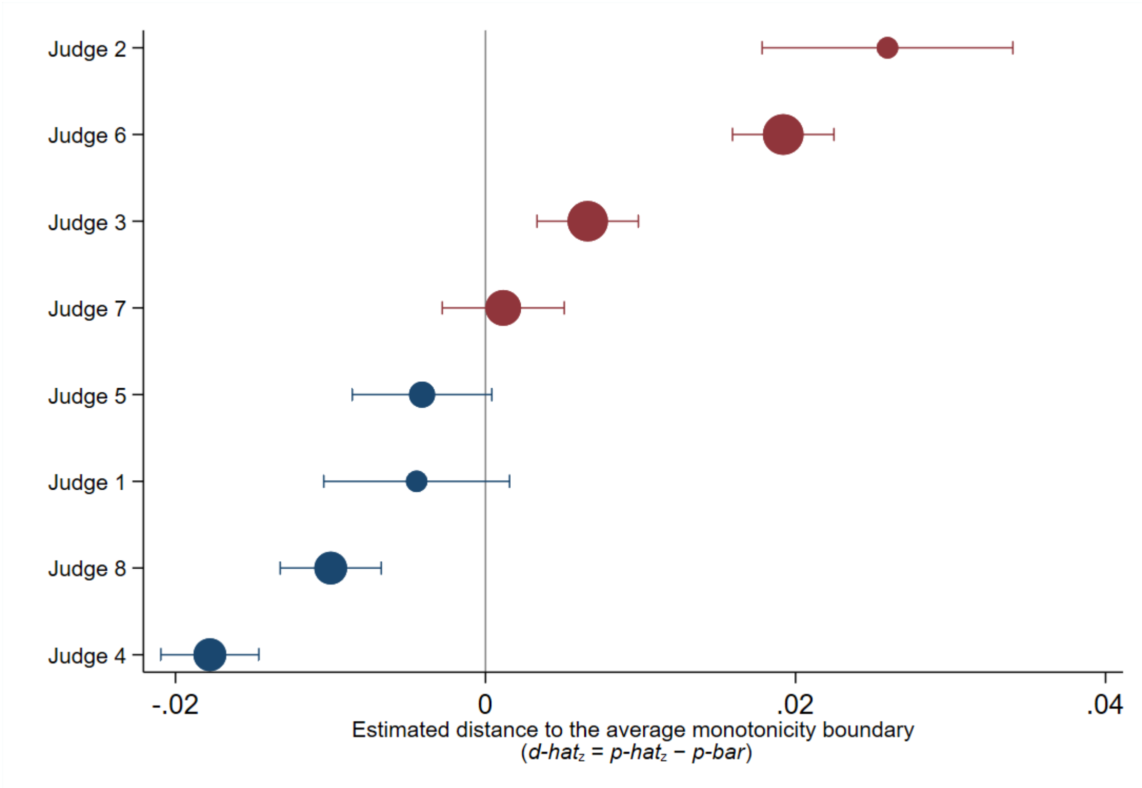

Figure 2: Judge Distances to the Average-Monotonicity Boundary in Philadelphia Bail

Notes: The figure plots $\hat{d}_z = \hat{p}_z - \hat{\bar{p}}$ for each judge, where $\hat{p}_z$ is the regression-adjusted treatment propensity and $\hat{\bar{p}}$ is the assignment-weighted average propensity. Horizontal bars report 95 percent confidence intervals. The vertical line at zero marks the average-monotonicity boundary. Judges to the left of zero lie below the boundary; judges to the right lie above it. Point sizes are proportional to empirical assignment shares $\hat{\lambda}_z$.

5 Conclusion

Average monotonicity gives judge-IV estimands nonnegative weights, but it does not give them a fixed target population. The positive-weight response types switch discretely when a judge's treatment propensity crosses the assignment-weighted average, and when the switching types have positive population shares and distinct average treatment effects, that switch need not leave the IV estimand locally invariant. The estimand remains a proper weighted average of treatment effects; what average monotonicity does not guarantee is that the average is taken over a population pinned down by behavior alone.

The boundary diagnostic makes this stability checkable without testing average monotonicity itself. It classifies where each estimated judge propensity lies relative to the boundary and how precisely, but it does not measure how far the estimand would move across the populations the sample cannot separate, which we leave to future work. In the Philadelphia bail data, three judges handling about 29 percent of cases are boundary-fragile, even in a design earlier work does not reject. Researchers who invoke average monotonicity should report where estimated judge propensities lie relative to the assignment-weighted average, how precisely those distances are estimated, and how much case mass sits near the boundary. When boundary fragility is detected, a natural next step is to bound the estimand range across the two sign classifications for the fragile judges, or to appeal to partial identification as in Mogstad and Torgovitsky (2024); the extension to covariate-adjusted propensities, where the boundary becomes cell-specific, remains an open question.

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